Title

AI-Powered Personal Farming Assistant: SCI-Style Research Paper Draft

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Abstract

This paper presents an AI-powered farming assistant integrating machine learning, computer vision, NLP, and large language models to support crop selection, disease diagnosis, irrigation, fertilizer recommendation, weather awareness, and market insights.

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1. Introduction

Agriculture faces challenges including climate variability, pests, soil degradation, and limited access to expert advice. AI can provide personalized recommendations.

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2. Literature Review

Recent studies explore deep learning for crop disease detection, LLM-based advisory systems, RAG, and precision agriculture.

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3. Problem Statement

Farmers often lack timely, localized, and multilingual decision support.

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4. Objectives

Develop a unified AI assistant for crop advisory, disease detection, irrigation, fertilizer optimization, and market intelligence.

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5. Methodology

Collect agricultural datasets, train ML/CV models, integrate LLM + RAG, expose through web/mobile application.

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6. Proposed Architecture

User -> Frontend -> AI Agents -> ML Models + Knowledge Base -> Recommendations.

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7. Expected Results

Improved decision making, higher crop yield, reduced input costs, faster disease diagnosis.

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8. Future Scope

IoT integration, drone imagery, satellite analytics, autonomous farm agents.

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9. Conclusion

An agentic AI farming assistant can improve accessibility and productivity through intelligent recommendations.

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References

Include recent SCI/SCIE papers from 2025–2026 on AI in agriculture, computer vision, LLMs, and precision farming.

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